PRASHANTH KUMAR SIRUSALA

Data Engineer

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TECHNICAL SKILLS

Data Platforms & Data Warehousing: Snowflake, Amazon Redshift, PostgreSQL, SQL Server, Apache Iceberg

Data Engineering & Modeling: SQL, Python, PySpark, Spark, dbt, SSIS, Airflow/MWAA, ETL/ELT, CDC, incremental models, dimensional modeling, star schemas, window functions, data quality checks, reconciliation

AWS & Cloud: Glue, Lambda, Step Functions, S3, Redshift, Athena, Glue Data Catalog, EventBridge, SQS, SNS, DMS, Aurora PostgreSQL, IAM, KMS, CloudWatch, Secrets Manager, AWS CDK

Streaming & Integration: Amazon Kinesis, Apache Kafka, streaming pipelines, event-driven pipelines, batch pipelines, REST APIs, SaaS and API ingestion, watermarking patterns

Infrastructure, DevOps & Monitoring: Terraform, AWS CDK, Docker, GitHub Actions, CI/CD, Datadog, CloudWatch monitoring

GenAI / LLM Integration: Claude API, AI-assisted ETL coverage checks in CI (GitHub Actions + Lambda), Snowflake MCP servers, governed-data chatbot, Snowflake Cortex

Security, Governance & Analytics: Snowflake RBAC, AWS IAM, KMS encryption, least-privilege access, HIPAA PHI/PII handling, data masking, lineage, data freshness monitoring, Tableau, Salesforce Marketing Cloud

PROFESSIONAL EXPERIENCE

Data Engineer — Tech Detectives Inc. February 2025 – Present

- Cut query time 70–80% on a 17B-row Snowflake security-event workload: traced the bottleneck to repeated raw-table scans and semi-structured attribute parsing, then moved high-use attributes into incrementally refreshed relational tables.
- Modernized legacy SQL Server and SSIS reporting with CDC into S3, Apache Iceberg and Athena, orchestrated by Step Functions; the refresh window dropped from 18 hours to 2.
- Added AI-assisted ETL coverage checks to pull-request approvals: GitHub Actions calls AWS Lambda and the enterprise Claude API to trace raw-to-curated coverage and flag data-loss gaps before release.
- Built an event-driven file-processing architecture on AWS: SQS queues buffer incoming source files, Step Functions route each file to a source-specific Glue job, and a completion gate holds downstream fact loads until every required feed has landed.
- Set up Snowflake MCP servers and a business-facing chatbot over governed data, cutting ad hoc engineering requests and the turnaround on day-to-day data questions.
- Own AWS Glue, Step Functions and Snowflake pipelines across batch, streaming (Amazon Kinesis, Apache Kafka), event-driven, API and CDC workloads, with retries, dependency control and recoverable failure paths.
- Write the PySpark and SQL transformations across raw, staging and curated layers: incremental loads, fact and dimension modeling, data quality checks, reconciliation and business rules.
- Productionized Airflow/MWAA and dbt workflows for SaaS and API-fed pipelines with reusable landing and watermarking patterns, incremental models, source freshness checks and controlled promotion across environments.
- Assessed warehouse options for a 67-table Aurora PostgreSQL source and a 1 TB analytics footprint, comparing Redshift Serverless, Zero-ETL, DMS CDC and Snowflake against integration and reporting needs.

Data Engineer — Triangulum Insights January 2024 – February 2025

- Maintained Redshift reporting tables of 450M rows each with incremental stored procedures and sequenced loads, refreshing them without full reloads.
- Built the AWS platform for an advertising and consumer panel-data business on S3, Glue, Step Functions, Lambda and Redshift, carrying source data from landing to reporting-ready tables.
- Delivered curated Redshift datasets to Tableau users and supported 50+ Salesforce Marketing Cloud automations for scheduled refreshes, query activities and campaign sends.
- Turned reporting-team grain, source and refresh requirements into Redshift table designs and the Glue and Step Functions pipelines behind them.
- Diagnosed long-running Redshift procedures and workload-management pressure with system tables, then re-sequenced execution and materialized-view refreshes around real workload patterns.
- Applied business-key and ingestion-order deduplication so new partitions merge into the reporting set while the latest valid record wins.
- Handled source and pipeline failures by isolating malformed files, fixing schema drift and metadata issues, and reprocessing only the affected inputs.

SSIS Engineer — Mindtree January 2020 – May 2022

- Built SSIS batch integrations moving healthcare data from source systems through SQL Server raw and staging layers into standardized reporting tables and views.
- Wrote SQL Server stored procedures, joins and scheduled transformations for raw-to-published data movement.
- Validated nightly outputs with source-to-target row counts and reconciliation checks, chasing missing records, duplicates and type conflicts before publication.
- Turned a 10-year healthcare retention rule into repeatable archival workflows, including an AWS-based archive for historical data.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate · CompTIA Security+

EDUCATION

M.S. Cybersecurity — Florida International University

B.Tech Computer Science & Engineering — Rajiv Gandhi University of Knowledge Technologies